



A Project Report

On

Profit Prediction Using Multiple Linear Regression

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Course : CDA

Submitted By

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1. Introduction

This project develops a supervised machine learning model to predict company Profit from three numerical expenditure variables: Research and Development (RD) Spend, Administration expenditure, and Marketing Spend. The source dataset contains 50 company observations and five columns: RD_Spend, Administration, Marketing_Spend, State, and Profit. The final prediction model intentionally excludes State and uses only the three expenditure variables as requested by the project design.

Exploratory analysis shows that RD Spend has the strongest linear association with Profit in the dataset (Pearson correlation = 0.9729), followed by Marketing Spend (correlation = 0.7478). Administration has a comparatively weak positive correlation with Profit (0.2007).

A Multiple Linear Regression model was trained using an 80/20 train-test split with random_state=42. On the held-out test set, the model achieved an R^2 of 0.9001 (90.01%), an RMSE of 8,995.91, and an MAE of 6,979.15. These results indicate that the model explains a substantial proportion of the observed variation in Profit on the test data, while the error metrics quantify the remaining prediction error.

The model was also used to demonstrate predictions for two new expenditure combinations. For the first combination (RD Spend 21,892.92, Administration 81,910.77, Marketing Spend 164,270.70), the predicted Profit is 71,236.89. For the second combination, the predicted Profit is 71,040.76.

2. Problem Statement

Companies allocate money to different business functions such as research and development, administration, and marketing. Understanding how these expenditure levels relate to profitability can support planning and forecasting. The problem addressed in this project is to estimate a company's expected Profit from its expenditure levels using a regression-based machine learning model.

The task is a regression problem because the target variable, Profit, is continuous. The model does not classify companies into categories; instead, it estimates a numerical profit value.

3. Project Objectives

- Understand the structure and quality of the company expenditure dataset.
- Perform exploratory data analysis to identify distributions, relationships, and patterns.
- Measure correlations between the numerical variables and Profit.
- Select RD Spend, Administration, and Marketing Spend as prediction features.
- Train a Multiple Linear Regression model using a reproducible train-test split.
- Evaluate model performance using RMSE, MSE, MAE, and R^2 .
- Compare actual and predicted Profit values on unseen test observations.
- Generate Profit predictions for new expenditure inputs.
- Translate the statistical results into practical analytical insights and recommendations.

4. Dataset Description

Column	Description	Data Type	Project Role
RD_Spend	Research and Development expenditure	Numerical	Input feature
Administration	Administration expenditure	Numerical	Input feature
Marketing_Spend	Marketing expenditure	Numerical	Input feature
State	Company state	Categorical	Exploratory only; excluded from final model
Profit	Company profit	Numerical	Target variable

The original dataset contains three numerical expenditure measures, one categorical State field, and the continuous Profit target. The State field was retained for descriptive analysis but was not passed to the final regression model.

5. Dataset Overview

The dataset contains 50 rows and 5 columns. There are 3 unique states: New York, California, Florida. Each of the three expenditure predictors and Profit is stored as a numeric value, while State is categorical.

Measure	Value
Rows	50
Columns	5
Numerical columns	4
Categorical columns	1
States	3
Duplicate rows	0

5.1 Descriptive Statistics

Variable	Count	Mean	Std Dev	Min	25%	Median	75%	Max
RD_Spend	50	73,721.62	45,902.26	0.00	39,936.37	73,051.08	101,602.80	165,349.20
Administration	50	121,344.64	28,017.80	51,283.14	103,730.88	122,699.79	144,842.18	182,645.56
Marketing_Spend	50	211,025.10	122,290.31	0.00	129,300.13	212,716.24	299,469.08	471,784.10
Profit	50	112,012.64	40,306.18	14,681.40	90,138.90	107,978.19	139,765.98	192,261.83

6. Data Quality Analysis

Missing-value inspection found no missing values in any of the five columns. The dataset also contains no duplicate rows. This means the modeling workflow did not require imputation or duplicate removal.

Data Quality Check	Result
Missing values – RD_Spend	0
Missing values – Administration	0
Missing values – Marketing_Spend	0
Missing values – State	0
Missing values – Profit	0
Duplicate rows	0

The dataset includes legitimate zero expenditure observations. For example, some companies have RD Spend equal to zero and/or Marketing Spend equal to zero. These values were not automatically treated as missing because zero can represent an actual expenditure level in the dataset.

6.1 Data types

Column	Pandas Data Type
RD_Spend	float64
Administration	float64
Marketing_Spend	float64
State	object
Profit	float64

7. Data Cleaning & Preprocessing

7.1. Checking Missing Values

First, the dataset was checked for missing or null values using:

```
df.isnull().sum()
```

```
RD_Spend      0
Administration 0
Marketing_Spend 0
State          0
Profit         0
dtype: int64
```

7.1 Checking Missing Values

The analysis showed that there are no missing values in the dataset. Therefore, no missing-value imputation or deletion was required.

7.2 Checking Duplicate Records

Duplicate records were checked using:

```
[167]: df.duplicated().sum()           #checking duplicates is
```

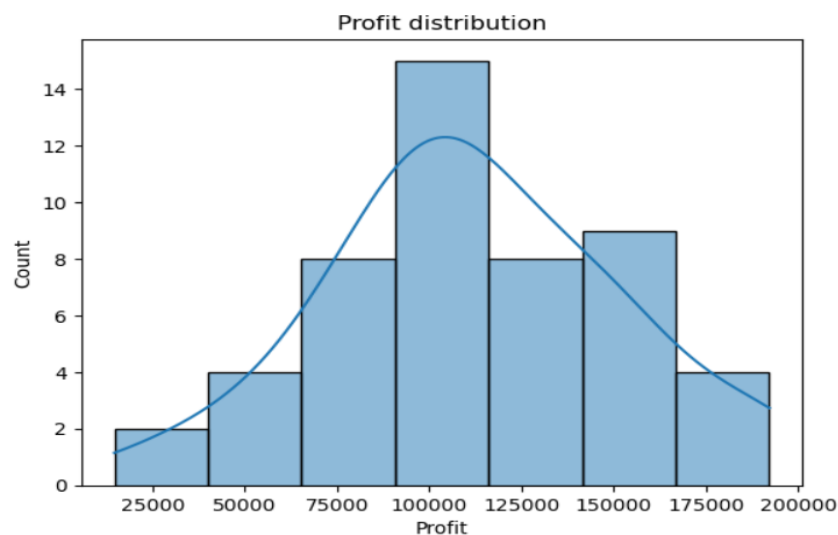
```
[167]: np.int64(0)
```

7.2 Checking Duplicate Records

The dataset contains no duplicate records.

8. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was used to understand the distribution of Profit and the relationships between the candidate predictors and Profit before interpreting the regression model.

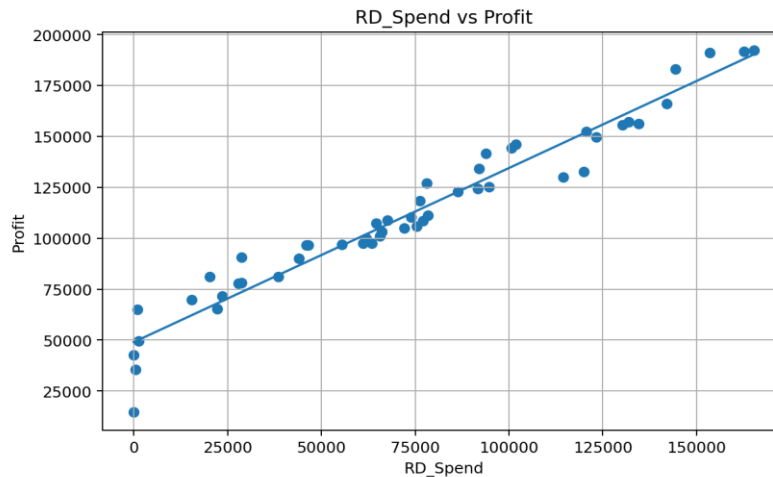


8. Profit Distribution

Profit ranges from 14,681.40 to 192,261.83, with a mean of 112,012.64 and a median of 107,978.19. The distribution therefore spans a wide range of profitability levels.

8.1 RD_Spend vs Profit

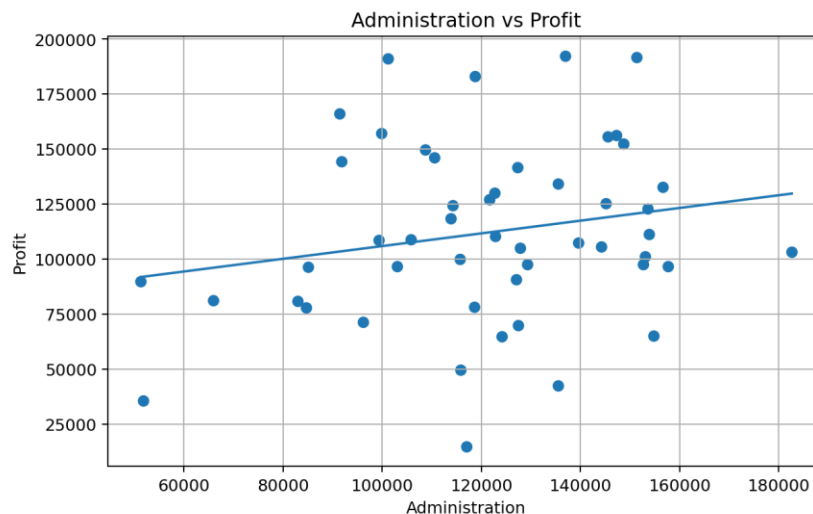
The Pearson correlation between RD_Spend and Profit is 0.9729, indicating a very strong positive linear association in this dataset. Correlation describes association and should not by itself be interpreted as proof that changing the expenditure causes the corresponding change in Profit.



8.1 RD Spend vs Profit

8.2 Administration vs Profit

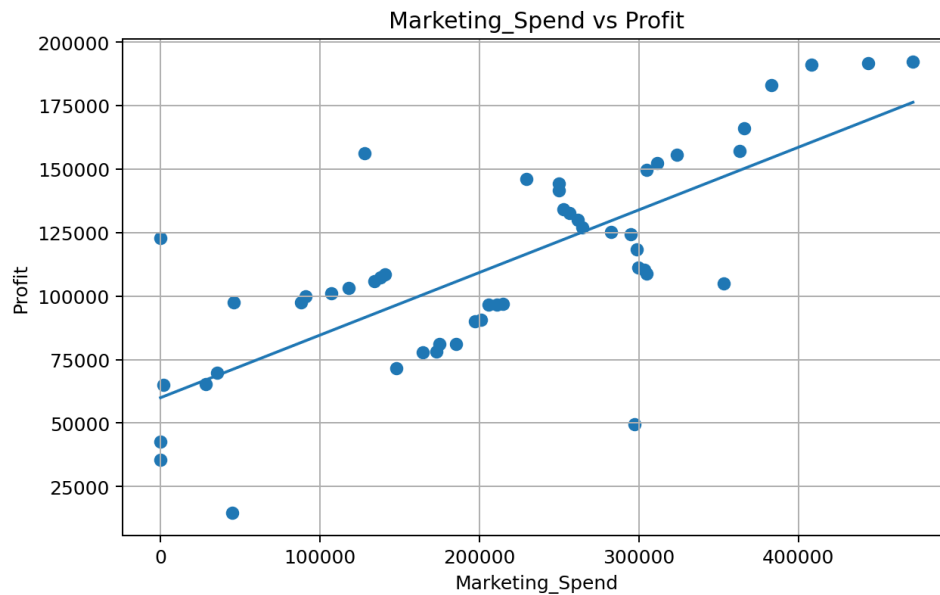
The Pearson correlation between Administration and Profit is 0.2007, indicating a weak positive linear association in this dataset. Correlation describes association and should not by itself be interpreted as proof that changing the expenditure causes the corresponding change in Profit.



8.2 Administration vs Profit

8.3 Marketing_Spend vs Profit

The Pearson correlation between Marketing_Spend and Profit is 0.7478, indicating a strong positive linear association in this dataset. Correlation describes association and should not by itself be interpreted as proof that changing the expenditure causes the corresponding change in Profit.

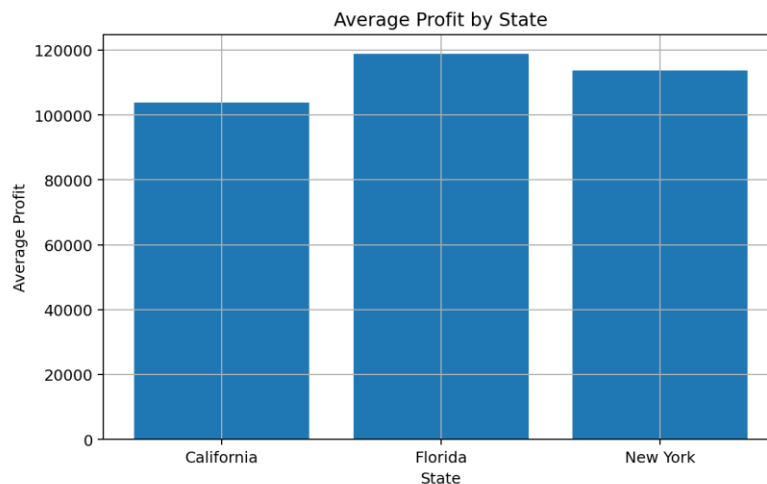


8.3 Marketing spend vs Profit

8.4 State Distribution and Descriptive Comparison

State was not included in the final predictive model, but it was reviewed descriptively. The dataset contains 17 California observations, 16 Florida observations, and 17 New York observations.

State	Number of Companies	Average Profit
New York	17	113,756.45
California	17	103,905.18
Florida	16	118,774.02



8.4 Average Profit by State

The state averages show differences in average Profit, but these differences are descriptive and should not be interpreted as a causal state effect. State was intentionally excluded from the final regression feature set.

9. Tools and Technologies

The following tools and technologies were used to develop and analyze the Profit Prediction model:

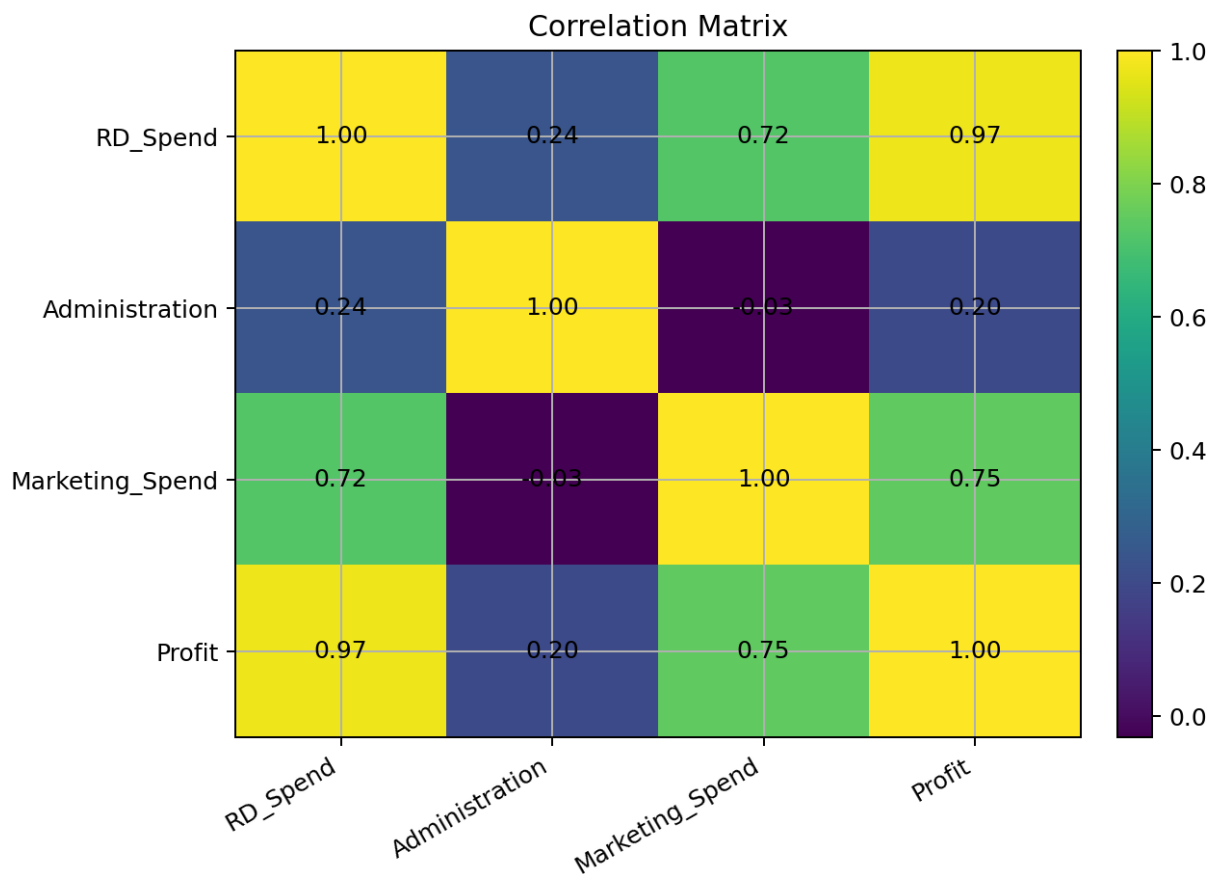
1. Python
2. Pandas
3. NumPy
4. Scikit-learn
5. Matplotlib
6. Power BI
7. CSV Dataset

10. Correlation Analysis

Correlation analysis quantifies the strength and direction of linear relationships among the numerical variables. The key relationships with Profit are shown below.

Feature	Correlation with Profit
RD_Spend	0.9729004656594831
Administration	0.20071656826872125
Marketing_Spend	0.7477657217414768

10. Correlation Analysis



RD Spend has the strongest relationship with Profit ($r=0.9729$). Marketing Spend also has a substantial positive relationship ($r=0.7478$). Administration has a much weaker relationship ($r=0.2007$). The predictors RD Spend and Marketing Spend are themselves correlated ($r=0.7242$), which is relevant when interpreting individual regression coefficients because predictor overlap can affect coefficient estimates.

11. Feature Selection

The target variable was separated from the predictors. The final feature matrix contains exactly three columns: RD_Spend, Administration, and Marketing_Spend. Profit is the target variable. State was explicitly excluded from X because the project requirement is to predict Profit from the three expenditure variables.

Variable	Used in Final Model?	Type
RD_Spend	Yes	Numerical
Administration	Yes	Numerical
Marketing_Spend	Yes	Numerical
State	No	Categorical
Profit	Target	Numerical

No categorical encoding was required because State was excluded. No missing-value imputation was required because the dataset contains no missing values. The final model therefore operates directly on the three numerical expenditure features.

12. Train-Test Split

The data was divided into training and testing subsets using `train_test_split` with `test_size=0.20` and `random_state=42`. With 50 total observations, this produces 40 training observations and 10 test observations. The test set is used as unseen data for evaluating the trained model.

Item	Value
Total observations	50
Training observations	40
Testing observations	10
Test proportion	20%
random_state	42

13. Multiple Linear Regression Model

Multiple Linear Regression models a continuous target as a linear combination of multiple explanatory variables. In this project, the model estimates Profit from RD Spend, Administration expenditure, and Marketing Spend.

The general model form is:

$$\text{Profit} = \beta_0 + \beta_1(\text{RD_Spend}) + \beta_2(\text{Administration}) + \beta_3(\text{Marketing_Spend})$$

Where :

β_0 = Intercept

β_1 = Coefficient of R&D Spend

β_2 = Coefficient of Administration Spend

β_3 = Coefficient of Marketing Spend

The fitted model coefficients are estimated from the training data using ordinary least squares. Each coefficient represents the estimated change in predicted Profit for a one-unit increase in that predictor while holding the other predictors constant.

14. Model Training and Coefficients

Model Parameter	Estimated Value
Intercept	54,071.88
RD_Spend	0.80
Administration	-0.07
Marketing_Spend	0.03

The fitted regression equation is: $\text{Profit} = 54,071.88 + (0.8038 \times \text{RD_Spend}) + (-0.0679 \times \text{Administration}) + (0.0312 \times \text{Marketing_Spend})$. The coefficients should be interpreted conditionally because the three predictors are modeled together.

In practical terms, the positive coefficient signs indicate that the fitted model increases predicted Profit as each expenditure variable increases, when the other model inputs are held constant. The numerical size of a coefficient is expressed in Profit units per one unit of the corresponding expenditure variable.

15. Model Evaluation

Metric	Value	Interpretation
R ²	0.900065	90.01% of test-set variance explained
MAE	6,979.15	Average absolute prediction error
MSE	80,926,321.22	Mean squared prediction error
RMSE	8,995.91	Typical error scale in Profit units

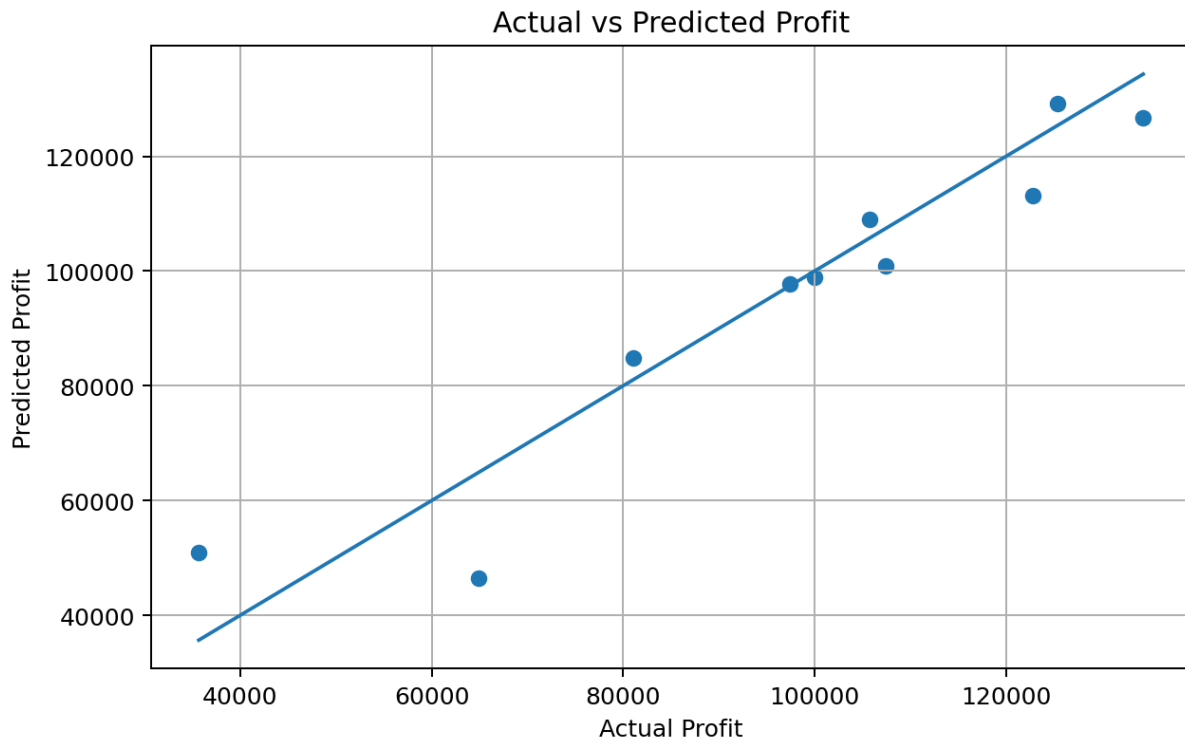
The test R² is 0.9001, meaning approximately 90.01% of the variation in the test-set Profit values is explained by the fitted linear model. The RMSE is 8,995.91, which provides an error measure in the same units as Profit. The MAE is 6,979.15, meaning the average absolute difference between actual and predicted Profit on the test observations is approximately 6,979.15.

These metrics indicate reasonably strong predictive performance for this small dataset, but the result should not be treated as a guarantee of future performance because the evaluation is based on only 10 test observations.

16. Actual vs Predicted Analysis

The test-set predictions were compared with the actual Profit values. The table below presents all ten held-out observations used in the evaluation.

Row Index	Actual Profit	Predicted Profit	Absolute Error
13	134,307.35	126,703.03	7,604.32
39	81,005.76	84,894.75	3,888.99
30	99,937.59	98,893.42	1,044.17
45	64,926.08	46,501.71	18,424.37
17	125,370.37	129,128.40	3,758.03
48	35,673.41	50,992.69	15,319.28
26	105,733.54	109,016.55	3,283.01
25	107,404.34	100,878.46	6,525.88
32	97,427.84	97,700.60	272.76
19	122,776.86	113,106.15	9,670.71



16. Actual vs Predicted Profit

Points close to the diagonal reference line indicate predictions close to the actual values. The plot therefore provides a visual assessment of how closely the model reproduces the observed test-set Profit values.

17. New Data Prediction

After training and evaluation, the model was used to estimate Profit for new expenditure combinations. The predictions below are directly from the trained model.

Scenario	RD Spend	Administration	Marketing Spend	Predicted Profit
Scenario 1	21,892.92	81,910.77	164,270.70	71,236.89
Scenario 2	23,940.93	96,489.63	137,001.10	71,040.76

For Scenario 1, the estimated Profit is 71,236.89. For Scenario 2, the estimated Profit is 71,040.76. These are model estimates based on the learned historical relationships and should be interpreted as forecasts rather than guaranteed financial outcomes.

18. Key Findings

- The dataset contains 50 observations with no missing values and no duplicate rows.
- Profit has a mean of 112,012.64, a median of 107,978.19, and a range from 14,681.40 to 192,261.83.
- RD Spend has the strongest linear correlation with Profit ($r=0.9729$).
- Marketing Spend has the second strongest positive correlation with Profit ($r=0.7478$).
- Administration has a relatively weak positive correlation with Profit ($r=0.2007$).
- The final regression model uses only RD Spend, Administration, and Marketing Spend, excluding State.
- The model achieved a test R^2 of 0.9001, with RMSE of 8,995.91 and MAE of 6,979.15.

- The actual-vs-predicted comparison shows that predictions generally track the direction and magnitude of the observed test-set Profit values, although individual errors remain.

19. Power BI Visualization

Power BI was used to visualize the dataset and understand the relationship between spending variables and Profit.

The following visualizations were created/recommended:

1. R&D Spend vs Profit :

A scatter chart was used to examine the relationship between R&D expenditure and Profit.

2. Marketing Spend vs Profit :

A scatter chart was used to examine the relationship between Marketing expenditure and Profit.

3. Administration Spend vs Profit :

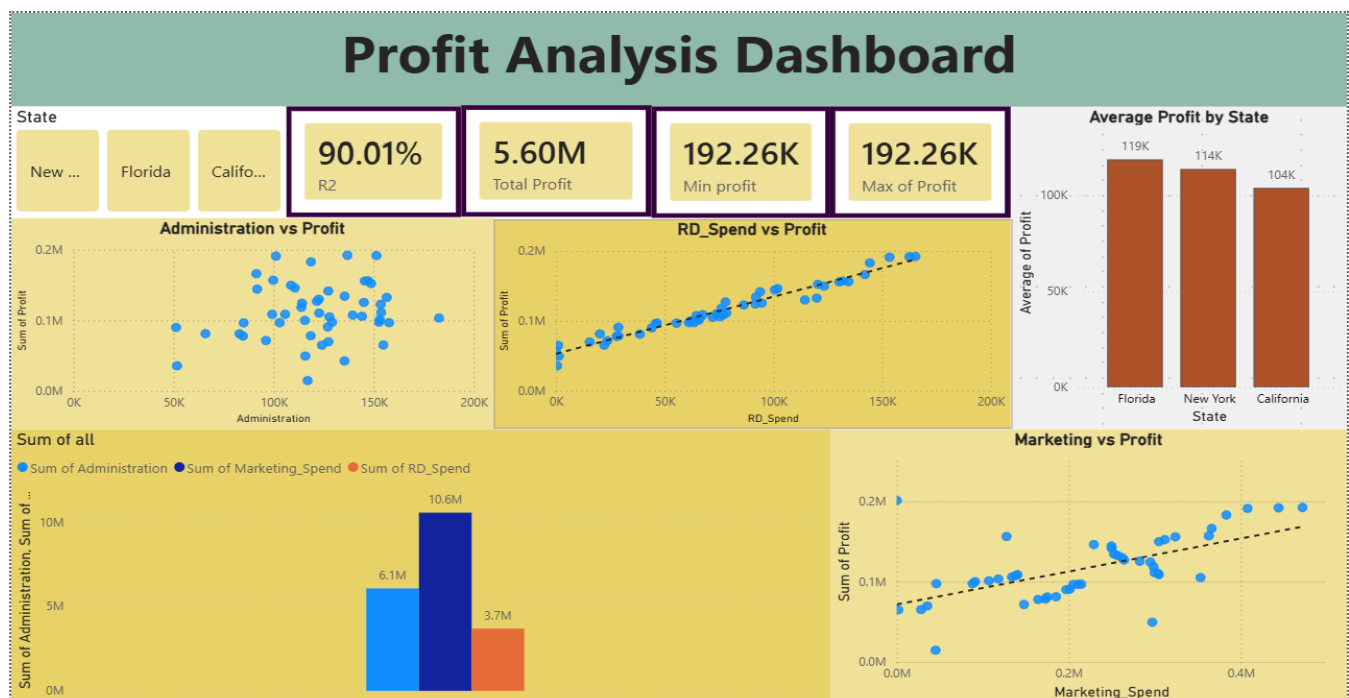
A scatter chart was used to examine the relationship between Administration expenditure and Profit.

4. State-wise Profit :

A bar or column chart was used to compare Profit across different States.

5. KPI Cards :

Important measures such as Total Profit, Average Profit, R&D Spend, and Marketing Spend were displayed using KPI cards.



19. Power BI Visualization

20. Business and Analytical Insights

Based on the regression analysis and Power BI visualizations, the following recommendations can be considered:

- The company should focus on the spending category that shows the strongest positive relationship with Profit.
- R&D expenditure should be monitored carefully because investment in Research and Development can be important for the long-term growth and profitability of a startup.
- Administration expenses should be reviewed regularly to ensure that unnecessary administrative costs are controlled.
- Marketing expenditure should be optimized according to its observed relationship with Profit instead of increasing the budget without measuring its effectiveness.
- Management should use data-driven analysis to decide how to allocate resources among R&D, Administration, and Marketing.
- Power BI dashboards can be used regularly to monitor Profit and spending patterns and support future business decisions.

21. Limitations

- The dataset contains only 50 observations, so statistical conclusions and model evaluation are limited by the small sample size.
- The model uses only three expenditure variables and does not include other potentially relevant business factors.
- State is excluded from the final model by project design, so the model cannot quantify any additional information associated with State.
- The model assumes a linear relationship between the predictors and Profit.
- The test evaluation is based on only 10 observations, so the reported performance can vary with a different train-test split.
- Correlation and regression association should not be interpreted as proof of causation.
- Predictions outside the historical range of the training data may be less reliable.

22. Conclusion

This project successfully developed a Multiple Linear Regression model for predicting company Profit from RD Spend, Administration expenditure, and Marketing Spend. The dataset was inspected for structure and quality, explored through descriptive statistics and visualizations, and analyzed using correlation and outlier screening. The final model was trained using 40 observations and evaluated on 10 unseen observations.

The resulting model achieved an R^2 of 0.9001, explaining approximately 90.01% of the variation in the test-set Profit values. The RMSE of 8,995.91 and MAE of 6,979.15 provide practical measures of the model's prediction error. The analysis indicates that RD Spend has the strongest simple linear relationship with Profit, while Marketing Spend also has a meaningful positive relationship.

The project demonstrates the complete basic machine learning workflow: data understanding, data quality checking, exploratory analysis, feature selection, train-test splitting, model training, evaluation, error analysis,

and prediction on new data. The resulting model can serve as the analytical engine for a future Python prediction program and Streamlit web application.

23. Future Scope

- Build a Streamlit interface so users can enter expenditure values and receive predicted Profit.
- Save the trained model as a .pkl file for reuse without retraining.
- Create a separate prediction Python program for deployment.
- Compare Linear Regression with Ridge, Lasso, Decision Tree, Random Forest, Gradient Boosting, and other suitable regressors.
- Use k-fold cross-validation and hyperparameter tuning for more robust evaluation.
- Add explainability features such as coefficient interpretation and feature importance where appropriate.
- Deploy the application to a cloud platform and monitor prediction performance.

24. Supporting Information

24.1 Core Modeling Code :

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

X = df[["RD_Spend", "Administration", "Marketing_Spend"]]
y = df["Profit"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)
```

25. Declaration

I hereby confirm that the analysis presented in this report was performed as part of the PRDA-01 Profit Analysis project. The regression analysis was performed using Python, and Power BI was used for data visualization and insight generation.

===== END =====